Viscosity & Terminal Velocity

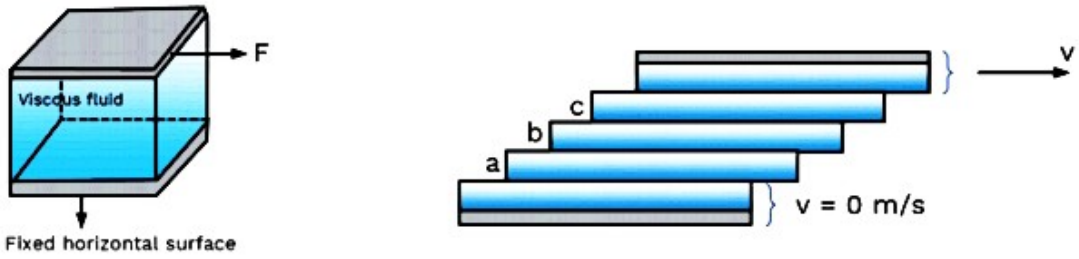
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Viscosity

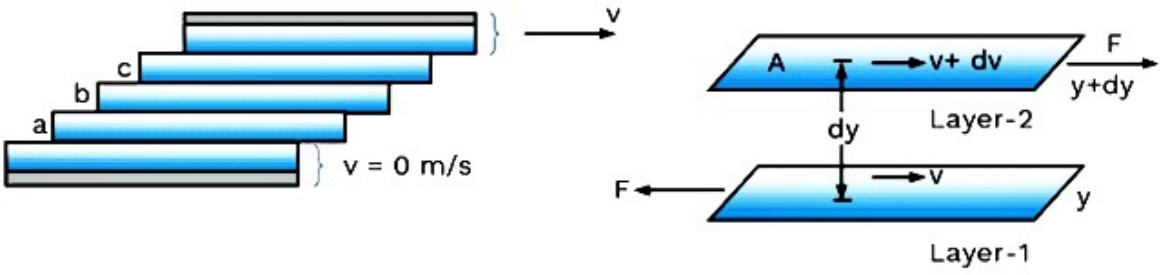
When a layer of a liquid slides over another layer of the same liquid, a frictional force acts between them which opposes the relative motion between the layers. This force is called 'internal frictional force' or 'viscosity'. Because of this property a backward dragging force (viscous drag) acts tangentially on the layers of the liquid in motion and it tries to destroy the motion of the liquid.



The layer of the liquid which is in contact with the surface is at rest, while the velocity of other layers increases with a distance from the fixed surface. Thus, there is a relative motion between adjacent layers of the liquid. Let us consider three parallel layers a, b and c, their velocities are in an increasing order. The layer a tends to retard the layer b, while b tends to retard c. If the flow of the liquid is to be maintained, an external force must be applied to overcome the dragging viscous forces. In the absence of the external force, the viscous forces would soon bring the liquid to rest. The property of the liquid by virtue of which it opposes the relative motion between its adjacent layers is known as 'viscosity'. Viscosity comes into play only when there is a relative motion between the layers of the same material. This is why it does not act in solids.

Measurement of viscosity

Consider a liquid flowing over a fixed horizontal solid surface in the form of parallel layers. The relative motion between the two layers can take place only, if some external force acts between them. Due to viscosity, a force F acts in opposite direction to destroy the relative motion.



According to Newton, the viscous force F depends upon the following factors:

- it is directly proportional to the area of the layers in contact i.e. $F \propto A$
- It is directly proportional to the velocity gradient between the layers i.e. $F \propto \frac{dv}{dy}$

  Combining these two factors, we have

$$F \propto A\frac{dv}{dy} \text{ or } F = -\eta A\frac{dv}{dy}$$

The negative sign indicates that the viscous drag acts in a direction opposite to the liquid flow

Here η is a constant of proportionality and is called the coefficient of viscosity or simply the viscosity of the liquid. Its value depends upon the physical conditions of the liquid.

Coefficient of viscosity of a liquid is defined as the tangential viscous drag acting per unit area between two liquid layers moving with unit velocity gradient.

$$\eta = \frac{(F / A)}{(dv / dy)} = \frac{\text{tangential stress}}{\text{velocity gradient}}$$

Also, we know that modulus of rigidity,

$$\eta = \frac{F / A}{\Delta x / L} = \frac{\text{tangential stress}}{\text{displacement gradient}}$$

Comparing equations we can observe that coefficient of viscosity in liquids is analogous to the modulus of rigidity in solids. Thus, viscosity of a liquid is the limiting case of rigidity in solids when they (i.e., solids) break down under the applied stress.

Dimensional formula for η

$$\eta = \frac{F}{A(dv/ dy)}$$

$$[\eta] = \frac{[MLT^{-2}]}{[L^2][LT^{-1}/ L]} = [ML^{-1}T^{-1}]$$

Units of coefficient of viscosity

(i) **In CGS system,** the unit of η is dyne s/cm^2 and is called poise (P).

(ii) **In SI System,** the unit of viscosity is Ns/m^2 and is called a pascal second (Pa s) as

$$1\ \text{Pa·s} = 1\frac{\text{N.s}}{\text{m}^2} = 1\left(\frac{10^5\,\text{dyne}}{10^4\,\text{cm}^2}\right).\text{s} = 10\left(\frac{\text{dyne.s}}{\text{cm}^2}\right)$$

$\Rightarrow$ 1 Pa.s = 10 poise = 1 decapoise

Effect of temperature on viscosity

Experiments show that η of a liquid decrease sharply with increase in its temperature and becomes zero at its boiling point, ON the other hand, η of gases increase with temperature.

$\eta \propto \frac{1}{\sqrt{T}}$ ..... for liquid

$\eta \propto \sqrt{T}$ ..... for gases

Q. **A plate of area 100 cm^2 is placed on the upper surface of castor oil, 2mm thick. Taking the coefficient of viscosity to be 15.5 poise, calculate the horizontal force necessary to move the plate with a velocity 3 cms^{-1}.**

Sol. The (horizontal tangential) viscous force is given by

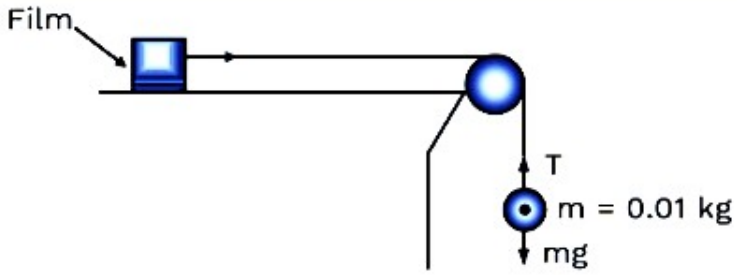
$$F = -\eta A\frac{dv}{dy}$$

Given η = 15.5 poise, A = 100 cm^2

$$\frac{dv}{dx} = \frac{v_2 - v_1}{x_2 - x_1} = \frac{(0-3)\,\text{cm/s}}{(2-0)\,\text{mm}} = -\frac{3}{0.2}\text{s}^{-1} = -15\,\text{s}^{-1}$$

$F = -15.5 \times 100 \times (-15) = 2.325 \times 10^4$dyne = 0.2325 N

Q. **A metal plate of area 0.10 m^2 is connected to a 0.01 kg mass via a string that passes over an ideal pulley (considered to be frictionless), as shown in the figure. A liquid with a film thickness of 3.0 mm is placed between the plate and the table. When released, the plate moves to the right with a constant speed of 0.085 ms^{-1}. Find the coefficient of viscosity of the liquid.**



Sol. As the metal plate moves with constant velocity,

$mg - T = 0 \Rightarrow T = mg$

If F is the viscous force on the plate, then.

$F = T = mg = 0.01 \times 9.8 = 9.8 \times 10^{-2} N$

By Newton's law of viscosity, $F = \eta A \frac{dv}{dy}$

$$\text{where } \frac{dv}{dy} = \frac{0.085}{3 \times 10^{-3}} = 28.3 s^{-1}$$

$\Rightarrow \quad 9.8 \times 10^{-2} = \eta \times 0.10 \times 28.3$

$\Rightarrow \quad \eta = 3.4 \times 10^{-2} N\ s/m^2$

Stokes' law

If you move a spherical body through a fluid, it experiences a viscous force. An expression for the viscous force experienced by the spherical object can be derived by the method of dimensional analysis.

Thus,

$F = k\eta^x v^y r^z$

$[MLT^{-2}] = [ML^{-1}T^{-1}]^x\ [LT^{-1}]^y [L]^z$

Equating indices of M, L ad T on both sides, we have

$1 = x$

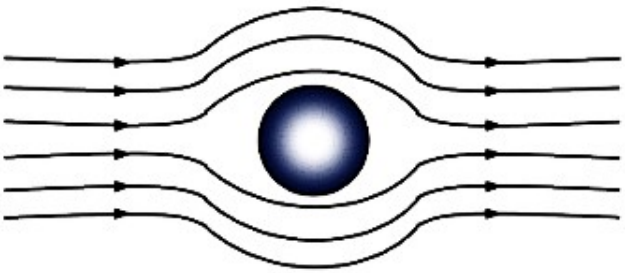
$1 = -x + y + z$

$-2 = -x - y$

Solving, we get

$x = 1, y = 1, z = 1$ and $k = 6\pi$ derived by Stokes

$F = 6\pi\eta rv$ ..... ***Stokes' law***



The Stokes' law is valid under the following assumptions:

(i) The medium through which the body moves is of infinite extent.

(ii) The moving body is perfectly smooth and rigid.

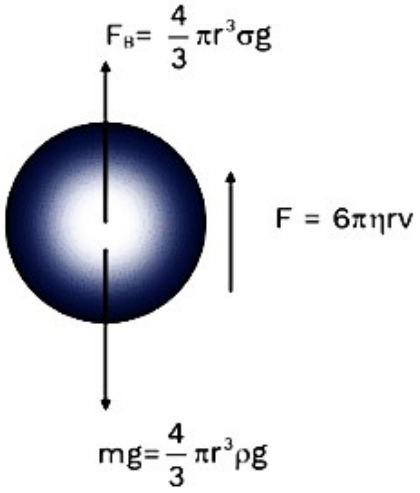
(iii) Though the size of the moving body is small, it is much larger than the distance between the molecules of the medium and as such the medium may be considered as continuous.

(iv) There is no slip between the moving body and the medium.

(v) The velocity of the moving body is less than the critical velocity of the medium and as such the motion is streamlined.

Terminal velocity

When a body is dropped in a viscous fluid. it is first accelerated and then its acceleration becomes zero and it attains a constant velocity called terminal velocity. As the body falls, the following three forces act on the body [see figure]



(i) Weight of the body acting vertically downwards. It is given by

$$mg = \frac{4}{3}\pi r^3 \rho g$$

(ii) Upward thrust equal to the weight of the fluid displaced acting vertically upwards. It is given by

$$F_B = \frac{4}{3}\pi r^3 \sigma g$$

Since velocity is constant

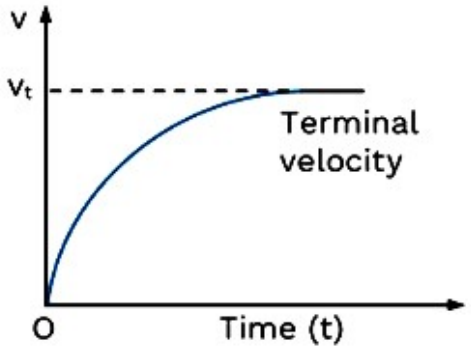
$\Rightarrow$ $F = 0$

$\Rightarrow$ $F_B + F = mg$

$$\Rightarrow \quad 6\pi\eta rv + \frac{4}{3}\pi r^3\sigma g = \frac{4}{3}\pi r^3\rho g$$

$$\Rightarrow \quad v = \frac{2r^2(\rho - \sigma)g}{9\eta} \qquad \text{.....Terminal velocity}$$

If we plot variation of velocity of the falling sphere with time we obtain a graph as shown in the figure.



Q. **Spherical particles of pollen are shaken up in water and allowed to settle. The depth of water is 2×10^{-2} m. What is the diameter of the largest particles remaining in suspension one hour later? Density of pollen = 1.8×10^3 kg m^{-3}, viscosity of water = 1×10^{-2} poise and density of water 1×10^{-5} kg m^{-3}.**

Sol. Terminal velocity, $v = \frac{2r^2}{9}\frac{(\rho - \sigma)g}{\eta}$

But we know $v = \frac{s}{t}$

$$\therefore \quad \frac{s}{t} = \frac{2}{9}\frac{r^2(\rho - \sigma)g}{\eta}$$

$$\Rightarrow r^2 = \frac{9s}{2t}\frac{\eta}{(\rho - \sigma)g}$$

Substituting given values, we get

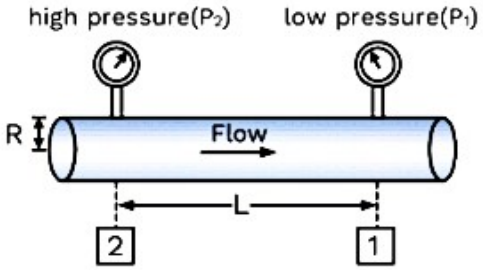
$$r^2 = \frac{9}{2} \times \frac{2 \times 10^{-2}}{3600} \times \frac{1 \times 10^{-3}}{(1.8 \times 10^3 - 1 \times 10^3) \times 10} = \frac{9}{36} \times \frac{1}{8} \times 10^{-10}$$

$$\Rightarrow \quad r = 1.77 \times 10^{-6} \text{ m}$$

Diameter D = 2r = 3.54 μm

Poiseuille's formula for flow of liquid through a capillary tube

When a liquid flows 'in a tube', viscous forces oppose the flow of the liquid. Hence a pressure difference is applied between the ends of the tube which maintains the flow of the liquid.



Poiseuille found that volume of liquid flowing per second (Q) is as follows:

Q is directly proportional to $P_2 - P_1 = \Delta P$, i.e $Q \propto \Delta P$

Q is inversely proportional to the length L, i.e. $Q \propto \frac{1}{L}$

Q is inversely proportional to the viscosity η, i.e. $Q \propto \frac{1}{\eta}$

Q is inversely proportional to fourth power of the radius, or R^4, i.e. $Q \propto R^4$.

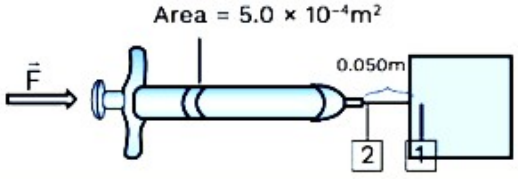
Combining all these factors, we have

$$Q \propto \frac{R^4 \Delta p}{\eta L} \text{ or } Q = \frac{kR^4 \Delta p}{\eta L}$$

Here, k is constant of proportionality and is found to be equal to $\pi/8$. therefore,

$$Q = \frac{\pi R^4 \Delta p}{8 \eta L} \quad \text{..... Poiseuille's equation}$$

Q. A hypodermic syringe is filled with a solution whose viscosity is 3.14×10^{-5} Pa-s. As figure shows, the plunger area of the syringe is 5.0×10^{-4} m², and the length of the needle is 5.0 cm. The internal radius of the needle is 2.0×10^{-4} m. The gauge pressure in the vein that is injected is 2000 Pa. What force must be applied to the plunger, so that 8.0×10^{-3} m³ of solution can be injected in 4.0 s?



Sol. We apply Poiseuille's law to the needle,

$$P_2 - P_1 = \frac{8\eta LQ}{\pi R^4} \qquad (P_1 \text{ is given gauge pressure})$$

The volume flow rate $Q = \frac{8.0 \times 10^{-6}\,m^3}{4.0s} = 2.0 \times 10^{-6}\,m^3/s$

Hence,

$$P_2 - P_1 = \frac{8\left(3.14 \times 10^{-5}\,Pa\,s\right)(0.05m)\left(2.0 \times 10^{-6}\,m^3/s\right)}{\pi\left(2.0 \times 10^{-4}\,m\right)^4} = 5000Pa$$

P_2 = 5000 Pa + 2000 Pa = 7000 Pa.

$F = (7000Pa)\left(5.0 \times 10^{-4}\,m^2\right) = 3.5N$

REYNOLDS NUMBER

Reynolds was able to show that flow of a liquid in a tube is laminar or turbulent depending upon the values of four physical parameters:

(i) average velocity of the liquid, v

(ii) viscosity of the liquid, η

(ii) density of the liquid, ρ and

(iv) diameter of the tube, d

In order to find a relation between them, let

$v \propto \eta^a \rho^b d^c$ or $v = k\eta^a \rho^b d^c$ .....(i)

Writing the dimensions of various quantities on both sides, we get,

$$\left[M^0 L^1 T^{-1}\right] = \left[ML^{-1}T^{-1}\right]^a \left[ML^{-3}\right]^b \left[L\right]^c$$

Equating the powers of M, L and T on both sides,

$a + b = 0$, $-a - 3b + c = 1$ and $-a = -1$

Solving these equations, we get

$a = 1$, $b = -1$ and $c = -1$

From Eq. (i),

$v = k\eta\rho^{-1}d^{-1}$ or $v = k\frac{\eta}{\rho d}$ .... Reynold formula

The dimensionless number k is called the Reynolds number and is denoted by R_e.

$$v = R_e \frac{\eta}{\rho d} \text{ or } R_e = \frac{\rho v d}{\eta}$$

Experiments have shown that if,

(i) $R_e < 2000$, liquid flow is laminar.

(ii) $R_e > 3000$, liquid flow is turbulent.

(iii) For values of R_e between 2000 and 3000, the flow is unstable and may change back ad forth from one type to the other.

The value of R_e when $v = v_c$ (critical velocity, the velocity of flow above which the flow becomes turbulent), is called the critical Reynolds number and is denoted by $(R_e)_c$.

$$v_c = (R_e)_c \frac{\eta}{\rho d} \text{ or } (R_e)_c = \frac{v_c \rho d}{\eta}$$

Q. **Calculate the Reynolds number for blood flowing at 30 cm/s through an aorta of radius 1.0 cm. Assume that the blood has a viscosity of 4 mPa-s and a density of 1060 kg/m³. What is the nature of flow?**

Sol. Here, $v = 30 \text{ cm/s} = 3 \times 10^{-1} \text{ m/s}$, $\rho = 1060 \text{ kg/m}^3$,

$d = 2 \times 1.0 \text{ cm} = 2 \times 10^{-2} \text{ m}$

$\eta = 4 \text{mPa s} = 4 \times 10^{-3} \text{ Pa s} = 4 \times 10^{-3} \text{ Ns/m}^2$

Thus, $R_e = \frac{v\rho d}{\eta}$

$$= \frac{\left(3 \times 10^{-1} \text{m/s}\right)\left(1060 \text{kg/m}^3\right)\left(2.0 \times 10^{-2} \text{m}\right)}{\left(4 \times 10^{-3} \text{Ns/m}^2\right)} = 1590$$

Since R_e is less than 2000, this flow will be laminar.